

TESTLEAF TRAINING

Playwright Inbuilt Locators & Use Cases

Playwright locators are the recommended way to find and interact with UI elements. They provide auto-waiting, retry-ability, and reliable interaction with dynamic web pages.

Locator	Syntax	Best Use Case
<code>getByRole()</code>	<code>page.getByRole()</code>	Buttons, links, checkboxes, headings, inputs using accessible role/name
<code>getByText()</code>	<code>page.getByText()</code>	Locate an element using visible text
<code>getByLabel()</code>	<code>page.getByLabel()</code>	Form fields associated with labels
<code>getByPlaceholder()</code>	<code>page.getByPlaceholder()</code>	Input fields identified by placeholder
<code>getByAltText()</code>	<code>page.getByAltText()</code>	Images using alt text
<code>getByTitle()</code>	<code>page.getByTitle()</code>	Elements having a title attribute
<code>getByTestId()</code>	<code>page.getByTestId()</code>	Elements with a dedicated test ID
<code>locator()</code>	<code>page.locator()</code>	CSS/XPath or complex/custom element selection
<code>frameLocator()</code>	<code>page.frameLocator()</code>	Elements inside an iframe

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1. getByRole()

Used to locate elements based on their accessible role.

Code

```
await page.getByRole('button', { name: 'Login' }).click();
```

Example

```
getByRole('button'), getByRole('link', { name: 'Home' }), getByRole('heading', { name: 'Dashboard' })
```

Use case

Use when the element has a meaningful accessibility role and name. Generally preferred because it resembles how a real user understands the UI.

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2. getByText()

Finds an element based on visible text.

Code

```
await page.getByText('Welcome to Salesforce').click();
```

Example

`getByText('Submit')` or `getByText('Welcome', { exact: true })`

Use case

Use when visible text is a reliable and stable identifier.

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3. getByLabel()

Useful for form controls associated with a label.

Code

```
await page.getByLabel('Username').fill('vineeth');
```

Example

HTML: <label for="username">Username</label> <input id="username">

Use case

Use when testing forms where labels are available.

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4. getByPlaceholder()

Finds input fields using their placeholder.

Code

```
await page.getByPlaceholder('Enter username').fill('vineeth');
```

Example

HTML: <input placeholder="Enter username">

Use case

Use when the input has a unique and stable placeholder.

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5. getByAltText()

Used mainly for images.

Code

```
await page.getByAltText('Testleaf Logo').click();
```

Example

HTML:

Use case

Use when images have meaningful alt text.

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6. getByTitle()

Finds elements using the title attribute.

Code

```
await page.getByTitle('Delete user').click();
```

Example

HTML: <button title="Delete user">Delete</button>

Use case

Use when the application provides a stable title attribute.

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7. getByTestId()

Used when the development team provides a dedicated test identifier.

Code

```
await page.getByTestId('login-button').click();
```

Example

HTML: <button data-testid="login-button">Login</button>

Use case

Use when you need a locator independent of UI text, styling, or DOM structure. Useful for stable automation frameworks.

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8. locator()

The generic locator API. It can use CSS selectors and XPath.

Code

```
await page.locator('#username').fill('vineeth');
await page.locator('.login-button').click();
await page.locator('input[name="email"]').fill('test@gmail.com');
await page.locator('//button[text()="Login"]').click();
```

Example

CSS: #username, .login-button, input[name='email'] | XPath: //button[text()='Login']

Use case

Use when preferred user-facing locators are not sufficient, or when a specific CSS/XPath selector is required. Prefer stable CSS over XPath when possible.

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9. frameLocator()

Used when the target element is inside an iframe.

Code

```
const frame = page.frameLocator('#payment-frame');  
await frame.getByLabel('Card Number').fill('123456789');
```

Example

```
page.frameLocator('#myFrame').getByRole('button', { name: 'Submit' }).click()
```

Use case

Use when the element exists inside an iframe.

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Recommended Locator Preference

RECOMMENDED LOCATOR PREFERENCE

**getByRole() → getByLabel() → getByPlaceholder() →
getByText() → getByTestId() → getByAltText()/getTitle() →
locator(CSS) → locator(XPath)**

Important: This is a practical preference order, not an absolute rule. The best locator is one that is unique, stable, readable, and representative of the UI contract.

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Interview Answer

INTERVIEW ANSWER

Playwright provides built-in locator APIs such as `getByRole`, `getByText`, `getByLabel`, `getByPlaceholder`, `getByAltText`, `getByTitle`, `getByTestId`, and the generic locator API. We use them to identify UI elements reliably. Among them, `getByRole` is generally preferred because it uses accessibility semantics and makes tests more user-centric. `locator()` is useful when CSS or XPath is required, while `frameLocator()` is used for elements inside iframes.